

Khushi Kumari

Third year Undergraduate

Department of Computer Science and Engineering | ✉ kkhushi21@iitk.ac.in | ☎ +91-7870988382 | in kkhushi21

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	BTech (CSE)	Indian Institute of Technology Kanpur	7.1/10.0
2021	Class XII (CBSE)	Scottish Central School, Sasaram	94.6%
2019	Class X (CBSE)	Scottish Central School, Sasaram	97.8%

Achievements

- Secured **All India Rank 2731** in **Joint Entrance Examination Advanced** among **0.2 Million** shortlisted students
- Secured **All India Rank 5158** in **Joint Entrance Examination Mains** among **1 Million** shortlisted students
- Achieved **State Rank 1** in class 9th in **National Level Science Talent Search Examination** conducted by **Unified Council**

Key Projects

Image Processing and Computer Vision Stamatics, IIT Kanpur May'23-July'23	
Image Processing & Application	- Learnt image enhancement and denoising techniques like: blurring, Thresholding, contrasting, filtering, edge detection, image segmentation, histogram computation and equalization - Performed rearrangements of the sub-matrices of an image to solve a jigsaw puzzle using matrix manipulation - Employed OpenCV's noise reduction algorithms to improve the quality of CCTV images and enhancing details for better object identification using histogram equalisation and contrast adjustments - Developed a program using OpenCV for the removal of shadows from gutters of scanned documents and applied techniques such as Adaptive thresholding, sharpening to improve visibility and clarity
Machine Learning	- Predicted the label of an image from a dataset consisting of 16000 grayscale images of documents - Developed a robust Deep Neural Network model for multiclass image classification using Tensorflow - Applied image enhancement techniques on the target data set to achieve optimal accuracy and generalization

Loan Repaid Prediction Self Project 🔄 July'23
- Imported the Loan dataset from Pandas library having 13 features and one label - Done some data analysis by plotting various plots like countplot, scatterplot to discover some correlation among the labels - Improved the accuracy from 0.73 to 0.84 by using Random Forest Classifier Model instead of Decision Tree Classifier

Technical Skills

Programming Languages	C, C++, Python
Exposure	Verilog HDL, MIPS Assembly Language, HTML, CSS, Javascript
Libraries	NumPy, Pandas, Matplotlib, Scikit-Learn, OpenCV
Software and Utilities	Github, Latex, Adobe Illustrator, Figma

Positions of Responsibility

Events & Design Secretary Skating Club, Games & Sports Council, IIT Kanpur Aug'22-Mar'23
- Managed various skating workshops organised by the club for the campus students along with other secretaries Designed posters for the circulation of important information regarding the club activities and workshops
Student Guide Counselling Service, IIT Kanpur Oct'22 - Present
- Ensured smooth transition of freshers into campus by personally guiding a group of 6 counselees academically and emotionally - Helped them with their academics by arranging meetings with mentors and still maintaining a good bond with them - Assisted Counselling Service in the conduction of week long UG Orientation Program 2022 of 1200 freshmen

Relevant Courses			*: Ongoing
Fundamental of Computing Mathematics for Computer Science III Introduction to Machine Learning*	Data Structure and Algorithm Introduction to Electronics Operating Systems(CS330A)*	Computer Organisation(CS220A) Linear Algebra & ODE Algorithms II (CS345A)*	

Extracurricular Activites

- Participated in **75 km walk event** organised by **Adventure's Club, IIT Kanpur** for commemorating 75 years of Independence and glorious history of India under government initiative **Azadi Ka Amrit Mahotsav**
- Contributed in **designing posters and illustrations** for technical fest, **Techkriti, IIT Kanpur**
- Volunteered in **Plantation Drive** organised by **Prakriti, IIT Kanpur** to promote greenery and to create awareness
- Contributed in the making of **Hall 6 website** by designing hero image and graphics for different pages of the website